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Smoking Consequences Questionnaire—Spanish

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A total of 212 Spanish smokers completed a Spanish version of a smoking questionnaire based on the Smoking Consequences Questionnaire—Adult (A. L. Copeland, T. H. Brandon, & E. P. Quinn, 1995) and a nicotine dependence (ND) measure. Confirmatory factor analysis results supported an a priori defined 8-factor structure. The results also indicated good internal consistency for the instrument and the scales derived from each factor. Positive outcome smoking expectancies scales were significantly and substantially associated with ND scores. Also, after controlling for the influence of ND, the authors found higher smoking expectancies in women than in men in (a) weight control, (b) craving reduction and addictiveness, and (c) negative-affect reduction. The results support the instrument's construct validity.

Drug effect expectancies is an important construct for current social learning theories of drug motivation (Brandon, Wetter, & Baker, 1996). In general, these theories predict that positive outcome expectancies contribute to ongoing drug use and drug relapse (e.g., Cox & Klinger, 1988; Hine, Summers, Tilleczek, & Lewko, 1997; Marlatt, 1985; Niaura et al., 1988). Most drug expectancies research has investigated the role alcohol expectancies play in alcohol consumption and the behavioral effects of alcohol. In general, alcohol expectancies are predictive of future drinking, discriminate between problem and nonproblem drinkers, influence frequency and amount of drinking, and appear to moderate treatment outcome (see Hittner, 1997; Jones & McMahon, 1996; Oei, Fergusson, & Lee, 1998). As we describe next, these results appear to generalize to smoking findings.

Much of the research on smoking outcome expectancies has involved the development and refinement of the Smoking Consequences Questionnaire (SCQ; Brandon & Baker, 1991; Copeland, Brandon, & Quinn, 1995; Wetter et al., 1994). The original version of the SCQ (Brandon & Baker, 1991) was developed with three goals in mind: (a) to measure the subjective expected utility (SEU) of smoking expectancies, (b) to discover the principal dimensions of smoking expectancies, and (c) to examine the relation between smoking expectancies and degree of smoking. The SEU is the crossproduct of the self-reported likelihood and the self-reported desirability of any given outcome.

Brandon and Baker (1991) administered an original pool of 80 items to smoker and nonsmoker undergraduate students. Applying exploratory factor analysis (EFA) to the data collected from the 382 smokers in their sample, Brandon and Baker identified 50 items that yielded four reliable factors or SEU scales: Negative Consequences (18 items), Positive Reinforcement–Sensory Satisfaction (15 items), Negative Reinforcement–Negative Affect Reduction (12 items), and Appetite–Weight Control (5 items). Brandon and Baker also found that women reported higher SEU scores than men on the Negative Reinforcement and the Appetite–Weight Control scales. Moreover, the results also showed that SCQ scores discrimi-

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nated between never-smokers, triers, occasional smokers, and regular smokers. However, contrary to expectations, likelihood scores were better than either desirability or SEU scores at differentiating between smoking categories.

Brandon and Baker (1991) cautioned that the mean smoking rate of their sample was rather low in comparison to adult smokers in the general population. They also recommended the use of likelihood scores over desirability and SEU scores. Therefore, Wetter et al. (1994) proceeded to crossvalidate the factor structure and discriminant and predictive validity of the 50-item questionnaire with 632 older, heavier smokers who were recruited from smoking cessation trials. Using only likelihood ratings, these researchers used confirmatory factor analysis (CFA) to test and compare four different factor models (one, two, three, and four factors). Wetter et al. concluded that the original four-factor model proposed by Brandon and Baker fit their data best. Also, Wetter et al. partially replicated the gender effects reported by Brandon and Baker; that is, women expected more Negative Reinforcement, Appetite–Weight Control, and Negative Consequences from smoking than did men.

Although Wetter et al. (1994) found that smoking expectancies were not associated with measures of nicotine dependence (ND; cf. Brandon & Baker, 1991), the individual scales were predictive of withdrawal symptoms, negative affect, and perceived stress in successful abstainers. Moreover, their results showed that low Negative Reinforcement and high Appetite–Weight Control expectancies were modestly predictive of smoking cessation success. This finding was impressive because, after controlling for the influence of dependence measures, smoking expectancies "were equivalent or superior to traditional nicotine dependence measures in predicting cessation success" (p. 809).

Copeland et al. (1995) published a third article on the factor structure of the SCQ. They hypothesized that the dimensions of smoking expectancies could differ between experienced and inexperienced smokers (see Brown, Goldman, Inn, & Anderson, 1980, for a similar argument about alcohol expectancies). Copeland et al. pointed out that although the SCQ's four-factor structure appeared to generalize from light (Brandon & Baker, 1991) to heavy smokers (Wetter et al., 1994), Wetter et al.'s (1994) study did not test whether a more complex structure would have been a better fit than the original four-factor model.

Thus, Copeland et al. (1995) administered the initial 80 smoking expectancy items from Brandon and Baker's (1991) study to a sample of 227 ongoing heavy smokers, 126 heavy smokers entering smoking cessation treatment, and 54 ex-smokers. They conducted EFA analyses of both SEU scores and likelihood scores and obtained similar results with both types of ratings. However, as in Brandon and Baker's study, the scales derived from the likelihood ratings fared better in validity tests than the SEU scores. Copeland et al.'s EFA of the likelihood scores resulted in the selection of 55 items that yielded 10 interpretable factors or scales: Negative Affect Reduction (9 items), Stimulation/State Enhancement (7 items), Health Risks (4 items), Taste/Sensorimotor (9 items), Social Facilitation (5 items), Weight Control (5 items), Craving/Addiction (6 items), Negative Physical Feelings (3 items), Boredom Reduction (4 items), and Negative Social Impression (3 items). Copeland et al. called their instrument the *SCQ—Adult* (SCQ–A).

Copeland et al. (1995) reported differences across smoking-status groups on four of the likelihood scales (Health Risks, Craving/Addiction, Negative Physical Feelings, and Negative Social Impression). They also reported gender differences in Health Risks, Weight Control, Negative Social Impression, and Taste/Sensorimotor expectancies, with women scoring higher than men in all of these but the Taste/Sensorimotor scale. Copeland et al. also reported significant positive correlations ranging from .20 to .12 between degree of nicotine addiction and 6 of the 10 scales: Negative Affect Reduction, Craving/Addiction, Stimulation/State Enhancement, Boredom Reduction, Social Facilitation, and Weight Control. Finally, the scales were also associated with treatment outcome measures. It is more notable that smokers in the treatment group had significant pre-to-post expectancies decreases in 7 scales, with abstaining participants typically showing the greater decreases.

Reasons for a Spanish Version of the SCQ

Cross-cultural investigations are important because human behavior and their controlling variables can be common (*emic,* or universal), different (*etic,* or culture specific), or have both emic and etic aspects across different cultures (Matsumoto, 1996). Therefore, cross-cultural research allows investigators to make inferences about the truth of human behavior and act accordingly. For example, a recent Surgeon General's report concluded that different patterns of tobacco use are often observed across racial–ethnic groups and that cultural psychosocial variables are thought to influence the initiation, maintenance, and cessation of smoking (U.S. Department of Health and Human Services [USDHHS], 1998). The report called for the development of "culturally appropriate, tobacco control programs" (p. 14) as well as for further research to investigate the psychological factors that may influence the initiation, maintenance, and cessation of tobacco use among different ethnic groups (USDHHS, 1998).

Spanish epidemiological studies over the last decade have consistently estimated that about 37% of Spaniards over the age of 16 smoke (Alonso et al., 1998; Ministerio de Sanidad y Consumo [MSC], 1999a; Reig Ferrer, 1989; Richart Martínez, Cabrero García, Sancho García, & Reig Ferrer, 1993; Rodríguez et al., 1987; Steptoe & Wardle, 1996). In Spain, a country of 40 million, patterns of smoking differ considerably by age and sex. The highest smoking prevalence rates are found among Spaniards of 24–44 years of age (52%), followed by the cohort between 16 and 24 years of age (40%). From 1987 to 1997, smoking rates for men decreased considerably (from 55% to 45%). In contrast, women were less likely to smoke in 1987 than in 1997 (from 23% to 27%). Moreover, some studies have estimated that, among those between the ages of 18 to 24, women are more likely to smoke than men (38% vs. 35%; Steptoe & Wardle, 1996).

Spanish health authorities have affirmed that the increase in young female smokers will have grave consequences in deaths due to cancer and cardiovascular and respiratory illnesses (MSC, 1999a). Thus, the Spanish ministry of health recently initiated a national campaign to inform the public about the risks of smoking. The objective is to emphasize the positive effects of being a nonsmoker and to promote respect for current smoking limitations in smoke-free, public places (MSC, 1999a). Similar efforts to curb tobacco consumption include recent mandates that prohibit smoking on commercial flights (MSC, 1999b) and in public buses and trains (MSC, 1999c).

We were able to find a few reports on smoking attitudes and expectancies in Spanish smokers, the subject of our research. Peralbo Uzquiano, Fraga Carou, and Méndez Paz (1994) asked high school and university students how they felt when they smoked and how they expected they would feel if they quit smoking. Over 55% of the participants endorsed feeling well and feeling relaxed when smoking, whereas less than 8% of the participants endorsed feeling dizzy or coughing from smoking. Having a bad taste in one's mouth was the negative effect most frequently endorsed (19%) and being able to concentrate better was the positive effect least frequently endorsed (12%). It is interesting that most of the participants (about 75%) also indicated that they would feel better (e.g., breathe better) if they became nonsmokers. However, a large number of students (about 35%) also reported that they would feel more anxiety and agitation and be less alert if they gave up cigarettes; that is, Peralbo Uzquiano et al. found that young Spanish smokers were more likely to endorse positive effects than negative effects of smoking and expected to feel physically better but psychologically worse if they were to give up smoking.

The expectancy that cigarette smoking will help people to relax and that stress and smoking rates are positively related is a highly consistent finding among Spanish smokers (e.g., Heras Tébar, García Sanchón, Hernández López, Ballestín, & Nebot, 1997; Reig Ferrer, Caruana Vañó, & Sánchez, 1993). Regarding reasons for quitting smoking, Becoña Iglesias and Vásquez González (1998) reported that health concerns (64%) and the fear of becoming seriously ill (19%) were the motives most frequently cited. Conversely, social reasons (pressure from family members or the work environment, concern about the impact of secondhand smoke) were rarely motives for giving up cigarettes (<5%).

About 30 million people of Latin American

or other Spanish descent live in the United States. Although cultural differences exist among different Hispanic subgroups, the majority of Hispanics speak Spanish and are Roman Catholic (see USDHHS, 1998). Among adults living in the United States, the smoking prevalence rates among Hispanic adults is 18.9%, compared to 25.9% for Europeans. Among Hispanics, women smoke considerably less than men—15.1% and 22.9%, respectively (USDHHS, 1998).

Investigations that have compared smoking expectancies between Hispanics and White non-Hispanics have found several significant group differences (B. V. Marín, Pérez-Stable, Marín, Sabogal, & Otero-Sabogal, 1990; G. Marín, Marín, Otero-Sabogal, Sabogal, & Pérez-Stable, 1989; G. Marín, Marín, Pérez-Stable, Sabogal, & Otero-Sabogal, 1995; G. Marín, Pérez-Stable, Otero-Sabogal, Sabogal, & Marín, 1989). For example, G. Marín et al. (1995) compared smoking expectancies between Hispanic (both Spanish and English speaking) and non-Hispanic Whites (English speaking). Non-Hispanics rated smoking as more pleasant and enjoyable than Hispanics, although, among Hispanics, women enjoyed smoking more than men. When rating motives and antecedents of smoking, Hispanics attributed less importance to relaxation and meal cues than non-Hispanics did. However, both groups gave similar importance to smoking caused by emotional and social factors. Regarding smoking consequences, Hispanics felt less controlled by their smoking habit and found smoking more damaging to the health of their own children than non-Hispanics did.

A shortcoming of the smoking-expectancies research conducted with Spanish-speaking populations in both Spain and the United States is that researchers have assessed specific expectancies with single-item measures (Becoña Iglesias & Vásquez González, 1998; B. V. Marín et al., 1990; G. Marín et al., 1995; G. Marín, Marín, et al., 1989; G. Marín, Pérez-Stable, et al., 1989; Peralbo Uzquiano et al., 1994). As Tiffany (1990, 1992) has noted, measuring variables with a single question can be problematic, because the reliability of any single item may be low or impossible to estimate. Single items are also likely to fall short of accurately capturing the multiple semantic representations of the variable of interest.

Thus, we set out to test whether the multidimensional nature of smoking expectancies as measured by the SCQ would replicate with a sample of Spanish smokers. Our objective was to develop a reliable and valid instrument for the measurement of smoking expectancies in Spanish students and, we hoped, other Spanish-speaking populations. Therefore, we translated the SCQ–A into Spanish and administered it to a sample of Spanish smokers. We then eliminated "bad" items and kept psychometrically sound items. Finally, on the basis of the final pool of items and the findings of Copeland et al. (1995), we tested an eight-factor model of smoking expectancies using CFA.

Given that Spaniards have not been exposed to antitobacco campaigns and smoking restrictions to the extent to which the U.S. public has (see MSC, 1999a, 1999b, 1999c), and noting that Spanish smokers rarely cite social reasons as a motive for quitting smoking (Becoña Iglesias & Vásquez González, 1998), we anticipated that the Negative Social Impression dimension of smoking expectancies would not be as relevant for Spanish smokers as for U.S. smokers. The finding that young Spanish smokers rarely report proximal, physiologically negative consequences from smoking, but are likely to say that they would feel better physically if they were to quit smoking (Peralbo Uzquiano et al., 1994), suggests that young Spaniards are nonetheless aware of the negative health consequences of smoking. Likewise, given that previous research has reported that in Spain smoking rates increase with levels of stress (Heras Tébar et al., 1997), and that Spanish smokers believe that smoking is relaxing but that quitting would make them less alert (Peralbo Uzquiano et al., 1994), we anticipated that the Negative Affect Reduction and the Stimulation/State Enhancement expectancy dimensions would replicate. Finally, as predicted by expectancy theory, we anticipated that smoking severity would be positively associated with expectancy levels; we also tested whether previous gender effects found with the SCQ generalized to Spanish students (e.g., higher Appetite–Weight Control expectancies in women).

Method

Participants and Procedures

We recruited 212 Spanish smokers (65% female) from public gathering places within the main campus of the University of Alicante (e.g., the smoking section of the university's cafeteria). The campus is located in the outskirts of Alicante, a city on the east coast of Spain. Participants were asked for 10–15 min of their time to complete an anonymous survey about smoking (only 12 individuals declined to participate). Participants did not receive any compensation. The participants were either college students (n = 180) or university employees (n = 32). The mean age of the participants was 22.50 years (SD = 4.97). Over 70% of the participants smoked 10 or more cigarettes daily (M = 12.97, SD = 7.45), had smoked regularly for 3 years or more (M = 5.51, SD = 4.97), and had made at least one previous "serious" quitting attempt (M = 1.58, SD = 1.59).

Measures

Smoking history. In addition to recording demographic information, we included a questionnaire similar to Fagerström's (1978) Tolerance Questionnaire to measure ND. ND scores were derived from the following six questions:

1. At times when you can't smoke or don't have cigarettes, do you feel a craving for one?
2. Is it tough to keep from smoking for more than a few hours?
3. When you are sick enough to stay in bed, do you still smoke?
4. How soon do you smoke your first cigarette after waking up in the morning?
5. About how many cigarettes do you smoke daily?
6. How old are you, and at what age did you start to smoke regularly?

Each affirmative answer to Items 1–3 added 1 point to the total ND score. On the basis of the lowest quartile cutoff score for Item 4 and the highest quartile cutoff scores for Items 5 and 6, lighting up within 30 min of waking up, smoking 20 or more cigarettes per day, and being a regular smoker for 6 or more years also contributed 1 point each to the total ND score. Thus ND scores ranged from 0 to 6.

Smoking Consequences Questionnaire—Spanish (SCQ–S). The SCQ–S is a descendant of the SCQ–A (Copeland et al., 1995). The SCQ–A was derived from a total of 80 statements that included 16 types of smoking consequences generated on the basis of research on smoking motivation and reports by smokers (see Brandon & Baker, 1991). Copeland et al. (1995) performed principal-component analyses on the items' likelihood scores, which ranged from 0 to 9. They retained 55 items that yielded 10 interpretable factors. Coefficient alpha reliabilities for the scales ranged from .83 to .96.

Antonio Cepeda-Benito, a Spanish native who has lived the United States for the last 17 years, translated the 55 items of the SCQ–A into Spanish. An experienced and fluent instructor of Spanish as a second language, a U.S. English-speaking native, then translated back into English the Spanish version of the SCQ–A. This U.S. translator then compared her back-translation with the original English questionnaire item by item. She identified discrepancies and, if necessary, adjustments to the Spanish translation were resolved by discussion between both translators. Finally, the questionnaire was revised one more time by Abilio Reig Ferrer, also a Spanish native, and administered to a small group of Spanish smokers, whose feedback was incorporated into the final Spanish translation.

Discrepancies between the original SCQ–A and the back-translation typically involved sentences or words whose semantic meaning would have changed or become awkward if they had been translated literally. For example, the item "cigarettes keep me from overeating" was translated to say "el fumar me ayuda a no comer demasiado," which was then back-translated as "smoking helps me not to overeat." Both translators concurred that the meanings of the two English sentences, "cigarettes keep me from overeating" and "smoking helps me to not overeat," were highly equivalent and the English to Spanish translation was preferable to a more literal translation (it would be odd to say in Spanish that cigarettes "keep you from" overeating). Feedback from Abilio Reig Ferrer and Spanish smokers involved minor "tuneup" adjustments such as changing the order of clauses ("me pica la garganta despues de fumar [my throat burns after smoking]" to "despues de fumar me pica la garganta") or saying things in a slightly different way ("creo que mi trabajo es mejor cuando fumo [I feel I do a better job when I'm smoking]" to "creo que hago un mejor trabajo cuando fumo"). We made these changes to improve the readability of the items.

Data Analysis

Development of the SCQ–S. As the initial step in item analysis, we examined Kaiser's (1974) measure of sampling adequacy (MSA) to determine if the data were appropriate for factor analyses and to see if any items needed to be eliminated. Values of less than .50 are considered "unacceptable," and those in the .50–.70 range are considered "miserable to mediocre"; ideally, the value should be above .80 or .90 (Kaiser, 1974). We then examined internal consistency for the entire instrument and each of the subscales proposed

by Copeland et al. (1995), as well as item–total correlations within the subscales. Third, we performed CFA, using LISREL 8 (Jöreskog & Sörbom, 1993) with the generalized least squares (GLS) method (for recommendations for GLS, see Anderson & Gerbing, 1988; Browne, 1982, 1984), and specified an eight-factor model (cf. Copeland et al., 1995), which we also compared with a one-factor model. For fit indices, we examined the normed fit index (NFI; Bentler & Bonett, 1980), the Tucker–Lewis index (TLI; see Marsh, Balia, & McDonald, 1988), the comparative fit index (CFI; Bentler, 1990), and the root mean square error of approximation (RMSEA; Browne & Cudeck, 1993). Values of the NFI, TLI, and CFI range from 0 to 1.00, with a value close to 1.00 indicating a better fit (e.g., Byrne, 1989; Mulaik et al., 1989). The TLI and CFI have been found to be unaffected by sample size (Bentler, 1990; Marsh et al., 1988). For the RMSEA, values of less than .05 are considered a close fit and less than .08 an adequate fit (Finch & West, 1997).

Validity analysis and gender comparisons. Several reports have indicated significant positive associations between levels of ND and smoking expectancies, as well as gender differences on SCQ–A scores (Brandon & Baker, 1991; Copeland, 1995; cf. Wetter et al., 1994). Thus, to examine the construct validity of the SCQ–S, we used each of the smoking-expectancies total-score scales as the dependent variables and performed a multivariate analysis of variance with sex as the independent variable and ND as a covariate (the assumption of equal regression slopes in the groups was met as preliminary analyses indicated the covariate and gender did not interact to predict the dependent variables; see Stevens, 1992). This analysis strategy also allowed us to examine the relation between smoking expectancies and gender while controlling for, and indexing the association between, ND and smoking expectancies.

Results

Development of the SCQ–S

We decided to delete one item—"Nicotine 'fits' can be controlled by smoking"—because this item was not answered by over 10% of the participants, and several participants indicated difficulties in comprehending the item. After deleting this item, 203 complete questionnaires remained available for further statistical analyses. The overall MSA for all 54 items was .81, with MSAs for the individual items ranging from .51 to .93. We thus deleted 4 items that had MSAs below .65 ("miserable to mediocre"). Across the SCQ-A–defined subscales, item–total correlations ranged from −.14 to .82. Ten items with item–total correlations less than .5 were deleted. The deletion criteria (low MSAs and low item–total correlations) resulted in the complete elimination of two 3-item scales: Negative Physical Feelings and Negative Social Impression. The item composition of the eight remaining scales changed as follows: (a) Negative Affect Reduction retained 8 of 9 items, (b) Stimulation/State Enhancement retained 5 of 7 items, (c) Health Risks retained 3 of 4 items, (d) Taste/Sensorimotor retained 7 of 9 items, (e) Social Facilitation retained 5 of 5 items, (f) Weight Control retained 5 of 5 items, (g) Craving/Addiction retained 3 of 6 items, and (h) Boredom Reduction retained 4 of 4 items. All item MSAs were higher than .70, and the overall MSA for the 40 retained items was .88, or "meritorious."

Thus, the initial 10-factor 55-item model became an 8-factor 40-item model, which we then examined with CFA. Although the absolute chi-square was statistically significant, $\chi^2(712, N = 203) = 863.31, p < .0001$, which occurs at any large sample size, the RMSEA was .032, suggesting a close fit. Values for other fit indices were as follows: NFI = .98, TLI = 1.0, and CFI = 1.0, also suggesting an excellent fit. There were 6 items with relatively high modification indexes (range = 5.00–7.39). However, in all of these instances these items loaded higher on their intended factors (all over .51) than they would have loaded on their nonintended factors (all values of the expected-change loading matrix were below .31). That is, the fit indices, as well as the individual item loadings within each subscale, suggested a fairly good overall fit (see Table 1). Moreover, a chi-square difference test comparing the 8- and 1-factor models was statistically significant, $\chi^2(28, N = 203) = 335.23, p < .0001$, suggesting that the 8-factor model fit better than the 1-factor model.

Interfactor correlations ranged from −.28 to .44 (see Table 2). To examine the discriminant validity of the multifactorial model, we also examined confidence intervals around the factor correlations (± 2 *SE*) to see if any contained 1.0 (see Anderson & Gerbing, 1988). None of these confidence intervals contained 1.0, supporting the discriminant validity of the model. For the final version of the SCQ–S, the overall alpha

Table 1
Scales (Alphas), Items, and Item–Factor Loadings of the Smoking Consequences Questionnaire—Spanish

Scale and item–factor loadings	α
Negative Affect Reduction (.94)	
Los cigarrillos me ayudan a calmar mis cabreos.	.89
Un cigarrillo me calma cuando estoy enfadado.	.86
Cuando estoy nervioso, un cigarrillo me tranquiliza.	.83
Cuando estoy preocupado un cigarrillo me ayuda.	.83
Si me siento irritable, un cigarrillo me relaja.	.79
Un cigarrillo me ayuda a que se me pasen los enfados.	.78
Cuando hay tensión; un cigarrillo ayuda a disiparla.	.76
Cuando estoy tenso un cigarrillo me relaja.	.72
Stimulation/State Enhancement (.81)	
El fumar me hace sentir más energético.	.69
Creo que hago un mejor trabajo cuando fumo.	.68
El fumar me ayuda mantener mi buen humor.	.67
Cuando estoy aburrido y cansado, los cigarrillos me dan energía.	.63
Me siento mejor fisicamente después de un cigarrillo.	.39
Health Risks (.80)	
El fumar es malo para mi salud.	.89
Cuanto más fumo más arriesgo mi salud.	.84
Fumando corro el riesgo de padecer del corazón y de contraer cancer del pulmón.	.81
Taste/Sensorimotor (.88)	
Me gusta el sabor del tabaco cuando estoy fumando.	.94
Mientras fumo disfruto de los sabores del cigarrillo.	.88
Yo disfruto del sabor del tabaco.	.81
Los cigarrillos me saben bien.	.78
Disfruto sintiendo un cigarrillo entre mis labios.	.65
Me gusta sentir el humo entrando por mi boca y pasando por mi garganta.	.54
Social Facilitation (.80)	
Cuando estoy entre fumadores, un cigarrillo me ayuda a sentirme parte del grupo.	.80
Cuando estoy de fiesta me lo paso mejor si fumo.	.56
Me desenvuelvo mejor con la gente con un cigarrillo.	.54
Fumando disfruto más de las conversaciones.	.44
Fumando disfruto más de la compañia de otros.	.43
Weight Control (.91)	
El fumar me ayuda a no comer demasiado.	.89
El fumar controla mi apetito.	.87
El fumar me quita el hambre.	.85
El fumar me ayuda a no engordar.	.84
El fumar me ayuda a mantener mi peso.	.80
Craving/Addiction (.72)	
El fumar satisface los deseos de consumir nicotina.	.77
Un cigarrillo elimina mi ansia de tabaco.	.71
Cuanto más fumo más dependo de la nicotina.	.59
Boredom Reduction (.88)	
Si no tengo nada que hacer, un cigarro ayuda a pasar el tiempo.	.93
Cuando estoy solo, un cigarro me ayuda a pasar el tiempo.	.84
Los cigarros son buenos para mater el aburrimiento.	.75
Cuando estoy aburrido y cansado un cigarrillo me ayuda mucho.	.66

was .92, and subscale alphas ranged from .72 to .94 (see Table 1).

Construct Validity

We found evidence of a significant association between level of ND and smoking expectancies. The overall multivariate effect for the covariate or ND was statistically significant, Wilks's lambda = .77, $F(8, 191) = 7.17$, $p < .001$. The multivariate effect size was .231. Within the present analyses, univariate βs are equivalent to partial correlations between each

Table 2
Interfactor Correlations (Loadings of the Phi Matrix) of the Smoking Consequences Questionnaire—Spanish

Factor	1	2	3	4	5	6	7	8
1. Negative Affect Reduction	—	.23	−.01	.12	.13	.13	.11	.29
2. Stimulation/State Enhancement		—	−.28	.11	.33	.44	.44	.22
3. Health Risks			—	.21	−.06	−.05	.11	.02
4. Taste/Sensorimotor				—	−.10	−.01	.20	.17
5. Social Facilitation					—	.05	.32	.11
6. Weight Control						—	.32	.24
7. Craving/Addiction							—	.02
8. Boredom Reduction								—

of the scale expectancies and ND, with gender as the controlled-for variable. Cohen (1988) characterized $r = .1$ as a small effect size, .3 as a medium one, and .5 as a large effect size. Univariate βs ranged in absolute values from .10 to .41. On the basis of Bonferroni alphas (.05/8 = .006), ND was a significant predictor of all smoking expectancies scales but Negative Affect Reduction, Health Risks, and Weight Control. Thus, as a whole, the statistically significant associations between smoking expectancies and ND could be characterized as medium (range = .21–.41; see Table 3).

Gender Effects

The gender effect analyses also replicated previous gender–smoking expectancies associations. The overall multivariate gender effect was statistically significant, Wilks's lambda = .90; $F(8, 191) = 2.68, p < .01$. The multivariate effect size was .101. On the basis of Bonferroni alphas (.05/8 = .006), women reported significantly greater Craving/Addiction, Negative Affect Reduction, and Weight Control consequences from smoking than men. Univariate effect sizes (η^2) for these statistically significant dependent variables (were .06, .03, and .05, respectively). Cohen (1988) characterized $\eta^2 = .01$ as a small effect size, .06 as a medium one, and .14 as a large effect size. Thus, we could characterize the statistically significant effects as medium. Conversely, η^2s for the nonsignificant effects were very small (range = .001–.01), which indicates that sample size was not the main factor contributing to lack of statistical significance in these analyses.

Table 3
Summary of Partial Correlations Between Smoking Consequences Scale—Spanish Scale Scores and Nicotine Dependence, With Gender as the Controlled-For Variable

Scale	Univariate β	p
Negative Affect Reduction	.12	.074
Stimulation/State Enhancement	.30	.0001*
Health Risks	−.10	.143
Taste/Sensorimotor	.26	.0001*
Social Facilitation	.22	.002*
Weight Control	.15	.033
Craving/Addiction	.40	.0001*
Boredom Reduction	.21	.003*

* $p < .006$ (Bonferroni adjustment = .05/8).

Discussion

We proposed to develop a multidimensional measure of smoking expectancies for use with Spanish smokers. Our efforts yielded promising results, with the SCQ–S being the first multidimensional smoking expectancies questionnaire in Spanish for which psychometric properties have been reported. Moreover, in addition to the overall excellent internal consistency of the SCQ-S scales, the good fit and discriminant-validity properties observed for the instrument support the notion that drug use expectancies have distinct dimensions rather than a single general construct (Brandon & Baker, 1991; Copeland et al., 1995; Wetter et al., 1994; cf. Leigh, 1989). That is, smoking expectancies appear to be multidimensional and rather similar across U.S. and Spanish smokers.

The factor structure of the SCQ–S supported all the proposed dimensions but the Negative Physical Feelings and Negative Social Impres-

sion factors, which we eliminated prior to conducting CFA because their items yielded poor MSAs and low scale item–total correlations. In our original pool of items, these two scales were composed of only three items each, whereas each of the other smoking expectancies scales had a minimum of five items. That is, the Negative Physical Feelings and Negative Social Impression factors may have not replicated simply because their domains were not adequately sampled.

There might be other reasons why we did not completely replicate the factor structure reported by Copeland et al. (1995). For example, whereas Copeland et al. administered the SCQ–A to heavy ongoing smokers, smokers in treatment, and even ex-smokers, our sample consisted of ongoing but mostly college student, moderate smokers. Compatible with the notion that drug-outcome expectancies become more complex with experience (Christiansen, Goldman, & Inn, 1982; Copeland et al., 1995), the participants in our sample may not have been as consistently aware of the negative physical and social consequences of smoking.

Contextual and cultural differences may have played a role in the discrepancy of results, especially in the failure to measure the negative social consequences of smoking. That is, Spanish smokers do not face the social censure that U.S. and other European smokers experience. Recently, Madrid's top health official made headline news when he refused to restrict cigarette smoking at a Spanish-sponsored World Congress on Health and Urban Environment. Among other reasons for defending his position, the Spanish official said "In Spain, we believe that smoking is a personal right." In response to this incident, a spokesperson for the World Health Organization officer said "We have made a lot of progress in Europe in trying to reduce cigarette smoking. Apparently, that message hasn't filtered down to Spain as yet" (see Susman, 1998, p. 1).

Our results also suggest that SCQ–S has construct validity. In particular, controlling for gender effects, Stimulation/State Enhancement, Taste/Sensorimotor, Social Facilitation, Craving/Addiction, and Boredom Reduction were all positively associated with ND. These results are highly similar to those previously reported with the SCQ (e.g., Brandon & Baker, 1991; Copeland et al., 1995; Downey & Kilbey, 1995) and are congruent with models of drug use that predict that positive drug expectancies increase drug consumption (e.g., Marlatt, 1985; Niaura et al., 1988).

Controlling for level of ND, we found that women expected greater Craving/Addiction, Negative Affect Reduction, and Weight Control from smoking than men did. These gender effects are also a replication of highly consistent findings (Brandon & Baker, 1991; Copeland et al., 1995; Wetter et al., 1994). Moreover, the effects are also congruent with findings from smoking cessation trials that show that women experience greater cigarette withdrawal symptoms (e.g., cigarette cravings, anxiety, irritability), indicate much more concern about weight gain after quitting, and report greater ND (Gritz, Nielsen, & Brooks, 1996; Hatsukami, Skoog, Allen, & Bliss, 1995; Klesges, Meyers, Klesges, & La Vasque, 1989).

In brief, using a sample of Spanish smokers, we replicated the finding that smoking expectancies are best conceptualized as a multidimensional construct, that these dimensions are differentially associated with ND levels, and that smoking expectancies are different for men and women. Our data confirmed 8 of the 10 smoking expectancies factors suggested by Copeland et al. (1995). Moreover, the patterns of associations between the expectancies scales with ND and with gender were congruent with theories of drug use and with previous research findings. Although we may not have had appropriate items to attempt to confirm the Negative Physical Feelings and Negative Social Impression factors, this shortcoming is ameliorated by the fact that Copeland et al. found poor construct validity for these two factors. These authors even suggested that although they chose to keep these two scales "for future research," other investigators could decide to exclude them from the instrument.

The SCQ–S appears to be appropriate for use with Spanish smokers. However, those using the instrument should not assume that it is equally valid for all Spanish-speaking individuals regardless of origin or nationality. Although we tried to avoid colloquialisms and believe that most Spanish-speaking individuals should be able to understand the items without problems, Spanish-speaking smokers from out-

side of Spain may express thoughts in slightly different ways. Thus, clinicians and researchers should pretest the SCQ–S with their populations of interest and, if necessary, adjust the wording of the items. Moreover, it would probably be useful to conduct future research testing the instrument with older, more experienced Spanish smokers and with Spanish speakers in other countries.

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